

Assignment 1

Course “New Generation Data Models and DBMSs”

A crowdfunding company plans to develop a platform where its 100,000 registered members can either launch or support fundraising campaigns. Each member is uniquely identified by a system-generated ID and is associated with an email address, first and last name, encrypted password, and contact information (street, city, postal code, and phone numbers). A member can act either as a **campaign creator** or a **backer (supporter)**, and many act as both.

Backers must provide their credit card details, which include the card company (Visa, Mastercard, or American Express), the card number, and its expiration date. Creators must supply verified financial information, including an IBAN, SWIFT code, and a government-issued proof-of-identity number.

The platform supports around 2,000,000 campaigns each year. Every campaign is created by a single member (the creator) and is uniquely identified by a system-assigned campaign ID. Each campaign is described by a title, category (e.g., Technology, Art, Games, Education), a detailed description, a funding goal, and the start and end date of the fundraising period.

Backers pledge (promise) money to campaigns they are interested in supporting. On average, each campaign receives between 200 and 3000 pledges (1000 is the average value). For each pledge, the system records the amount pledged, the time of the pledge, and a boolean number representing the actual transfer of money. When the campaign period ends, the system determines whether the campaign was successful (i.e., if the total amount pledged meets or exceeds the funding goal). If the campaign is successful, the pledged funds are transferred from the backers to the creator (we wish to record this transfer along with the date in which the transaction has been made).

After a successful campaign, both the creator and each of the backers can leave feedback. Feedback consists of a numerical rating (on a scale from 1 to 10) and a textual comment describing their experience with the transaction.

Please respond to the following questions:

1. Design a UML class diagram for the proposed domain. Identify all the assumptions from which the design relies on and the constraints that should be specified on this domain.
2. In the proposed UML class diagram you should report the following workload:
 - a. Identify the top 5 most competitive campaigns, i.e., those that received the highest number of pledges.
 - b. For each campaign creator, identify the category in which they achieved the highest average pledge amount.
 - c. For each month in the last year, identify the top 10 creators per month, i.e., the creators who completed the highest number of successful campaigns in that month.

To draw the UML class diagram, you can use draw.io: <https://app.diagrams.net/>

Please provide any description that support me in understanding your solution.